Statistical validation of TESS candidates

TRICERATOPS validation pipeline — Observer's Guide

with the graphical user interface

Pipeline `triceratops_validation.py` v4.1.1 GUI `triceratops_gui.py` v1.0

November 2026

Abstract

This guide is written for observers and collaborators who want to use the validation pipeline without reading its 150-page technical manual. Part I explains what TRICERATOPS computes, which false-positive scenarios it considers, what the pipeline adds around it, and how ground-based photometry enters the analysis. Part II explains the graphical user interface (GUI) with a worked example, the AAVSO campaign 942 on TIC 65344135, and gives the first TESS-only result for that star (4 November 2026). It ends with a complete reference of all options, the output files and common problems. The technical manual (`TRICERATOPS_Validation_Pipeline_Documentation_v4.1.1.pdf`) remains the reference for the methods and their tests.

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Part I

Background

Chapter 1

Why statistical validation?

A transit candidate is a periodic dip in a light curve. A planet crossing its star produces such a dip, but so do several other configurations:

- an eclipsing binary star (EB) whose eclipses are grazing or diluted;
- an eclipsing binary in the background, whose light is mixed with that of the target (blend);
- a planet or EB around a fainter star physically bound to the target;
- an eclipsing binary on a nearby star a few arcseconds away, inside the same 21″ TESS pixel (nearby eclipsing binary, NEB).

For many TESS candidates, especially faint stars and long periods, the planet's mass cannot be measured with radial velocities. *Statistical validation* then asks: given everything we know about the star, its surroundings and the shape of the light curve, how probable is it that the signal is a planet on the target?

What this pipeline does

`triceratops_validation.py` prepares the TESS data, checks the candidate with a series of diagnostic tests, and then calls the TRICERATOPS package (Giacalone et al. 2021) to compute the false-positive probabilities. Since version 4.1 it also uses ground-based light curves, for example from an AAVSO campaign, to rule out nearby stars and to test the colour dependence of the depth. The GUI `triceratops_gui.py` lets you set all of this up in a window, run it and look at the results.

Chapter 2

How TRICERATOPS works

2.1 Scenarios

TRICERATOPS (Tool for Rating Interesting Candidate Exoplanets and Reliability Analysis of Transits Originating from Proximate Stars) compares the observed transit with a set of astrophysical *scenarios*. Each scenario is a possible origin of the dip:

Scenario	**Meaning**
TP	Transiting planet on the target star.
EB	Eclipsing binary: the target itself is an eclipsing binary.
EBx2P	The same at twice the period: two nearly equal eclipses per orbit are seen as one "transit" at half the true period.
PTP, PEB, PEBx2P	The target has an unresolved, physically bound companion; the planet or EB is on the *primary* (the target) and the companion dilutes the dip.
STP, SEB, SEBx2P	The planet or EB is on the unresolved bound *secondary* star.
DTP, DEB, DEBx2P	Planet or EB on the target, diluted by an unresolved chance-aligned (*diluting*) background or foreground star.
BTP, BEB, BEBx2P	Planet or EB on that unresolved *background* star.
NTP, NEB, NEBx2P	Planet or EB on a *nearby* star that is in the TIC and lies within the TESS aperture region (resolved by Gaia, not by TESS).

Only TP, PTP and DTP put a planet on the target. All other scenarios are false positives.

2.2 Probabilities

For each scenario S, TRICERATOPS computes a probability proportional to

$$P(S \mid \text{data}) \;\propto\; p(S) \int L(\text{data} \mid \theta, S)\, p(\theta \mid S)\, d\theta.$$

Prior $p(S)$. How common the configuration is:

- the occurrence rates of planets as a function of period and size;
- the statistics of stellar companions, as a function of separation and mass ratio;

- the density of background stars from a synthetic model of the Galaxy (TRILEGAL) for the area of the aperture.

Likelihood L. How well a light-curve model of that scenario fits the folded TESS light curve. The models include limb darkening and the dilution by the other stars in the aperture.

Integral. It runs over the unknown parameters θ: radius of the planet or companion star, impact parameter, eccentricity, the properties of the host star, and so on. It is evaluated by Monte Carlo: many random draws of θ, typically a million, for which the model light curves are compared with the data.

The results are summarised in two numbers:

$$\mathrm{FPP} = 1 - (P_{\mathrm{TP}} + P_{\mathrm{PTP}} + P_{\mathrm{DTP}}), \qquad \mathrm{NFPP} = \sum P_{\mathrm{NTP,\,NEB,\,NEBx2P}}.$$

FPP is the probability that the signal is *not* a planet on the target. NFPP is the part of it due to resolved nearby stars.

2.3 When is a candidate "validated"?

Following Giacalone et al. (2021), the pipeline reports **PASSED** when

$$\mathrm{FPP} < 0.015 \quad \text{and} \quad \mathrm{NFPP} < 0.001.$$

Other combinations mean:

- FPP < 0.5 with NFPP < 0.001: a *likely planet*;
- NFPP > 0.1: a likely nearby false positive.

A "PASSED" is a statistical statement. It is meaningful only when the diagnostic tests below show nothing suspicious, and it is no substitute for follow-up observations. These criteria were designed together with TESS follow-up practice: ground-based photometry to clear nearby stars, high-resolution imaging (contrast curves) against close companions, and reconnaissance spectroscopy.

2.4 What lowers the false-positive probability

A clean transit shape. A flat-bottomed U-shaped transit of the right duration for the star fits TP well and EB poorly. A V-shaped (grazing) dip does the opposite.

A precise depth and a correct dilution. The depth determines the radius of the occulting body. If the dilution by neighbours is wrong, the radius is wrong.

High-resolution imaging. A contrast curve (`--contrast-curve`) limits the brightness of unresolved companions and background stars close to the target. It lowers the P*, S*, D* and B* scenarios.

Ground-based photometry. A light curve that resolves the field shows *which* star dims. Stars shown not to vary are removed from the analysis, which removes their NTP/NEB scenarios and usually brings NFPP close to zero (Chapter 4). With TRICERATOPS+ the multi-colour light curves also enter the likelihood.

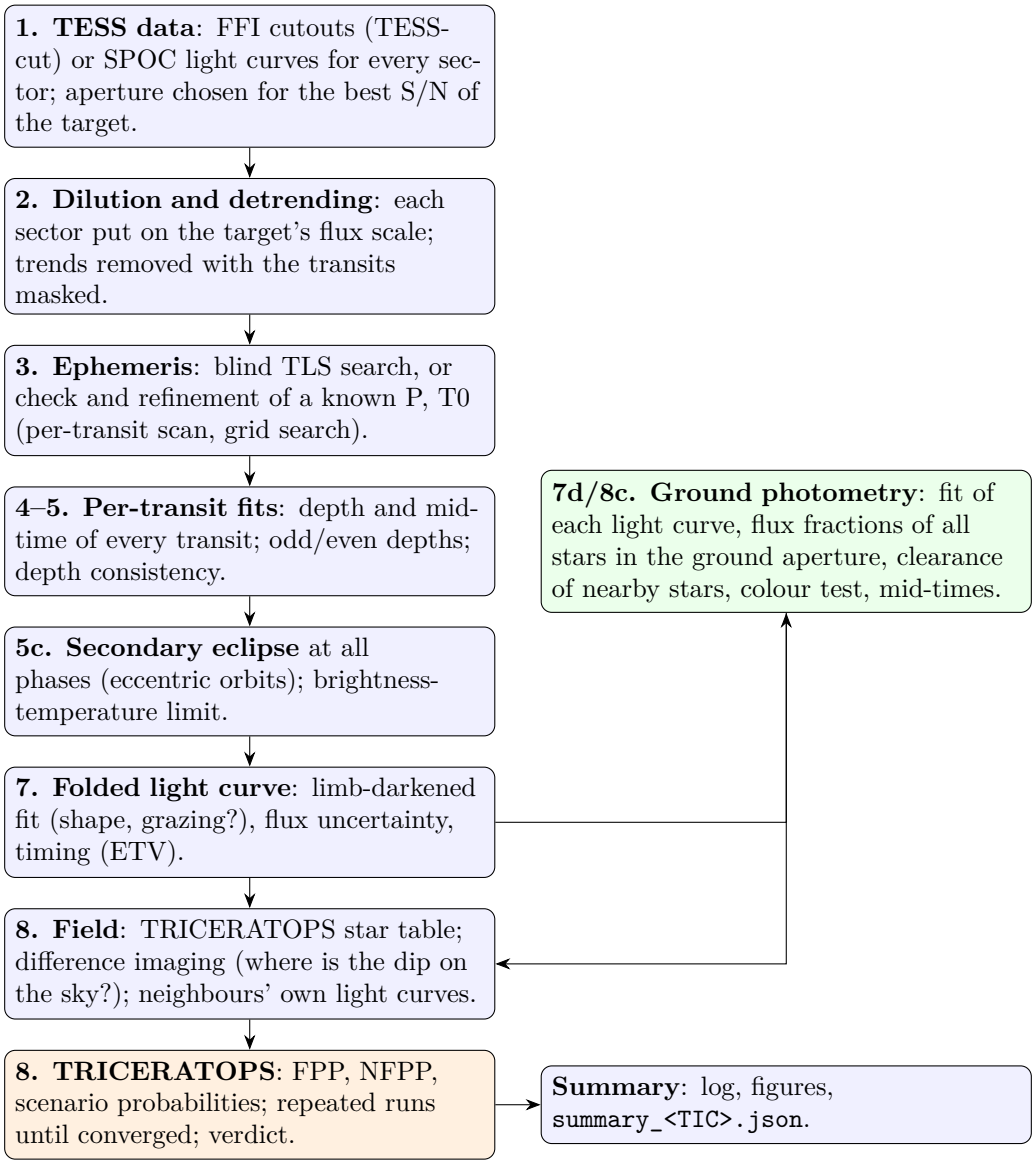
2.5 Limitations

- The Monte Carlo has noise. One run can be dominated by a lucky draw. The pipeline therefore repeats `calc_probs`, pools the evidences and adds runs until no single run dominates.
- The priors assume "typical" stars and planets; unusual systems are not well represented.
- With only a few transits, a long period or a shallow depth, the folded light curve constrains the shape poorly, and the FPP becomes uncertain.
- An FPP close to 1 with an EB scenario on top, as for TIC 229788204, is a strong indication of a stellar eclipse, but radial velocities are needed to measure the masses.

Chapter 3

What the pipeline does around TRICERATOPS

TRICERATOPS needs a good folded light curve, a correct depth and the pixels of the aperture. The pipeline prepares these and runs independent diagnostic tests:



The diagnostic tests and what they reveal:

Test	**What a failure indicates**
Odd/even depths	Different depths of alternating transits: an EB at twice the period.
Secondary eclipse	A dip elsewhere in the orbit: light from a self-luminous companion, so an EB. The limit gives a maximum temperature of the companion.
Limb-darkened fit	V shape, $b + k > 1$ (grazing), or a stellar density inconsistent with the star: EB or wrong host.
Difference imaging	The dip is centred on a neighbour instead of the target: NEB.
Neighbour vetting	A neighbour's own TESS light curve shows the dip more deeply.
Per-transit depths	Depths that differ between sectors: wrong dilution or a blend.
Timing (ETV)	Mid-time variations: a third body, or a wrong period.
Ground photometry	The dip on the "wrong" star, a wrong depth, or a colour-dependent depth (Chapter 4).

Chapter 4

Ground-based observations in the analysis

4.1 The blending problem

A TESS pixel is 21″ wide and the TESS point-spread function covers a few pixels. Stars a few arcseconds apart are fully blended. In TIC 65344135 the target ($V = 12.98$) and its neighbour TIC 65344134 ($V = 13.17$, at 5.9″) end up in the same pixel (Figure 4.1). TESS measures a dip of 3.2 ppt in their combined light. If the target is the host, its own dip must be about 5.9 ppt; if the neighbour is the host, about 7.0 ppt. Under 2–3″ seeing a ground-based telescope measures both stars separately and decides.

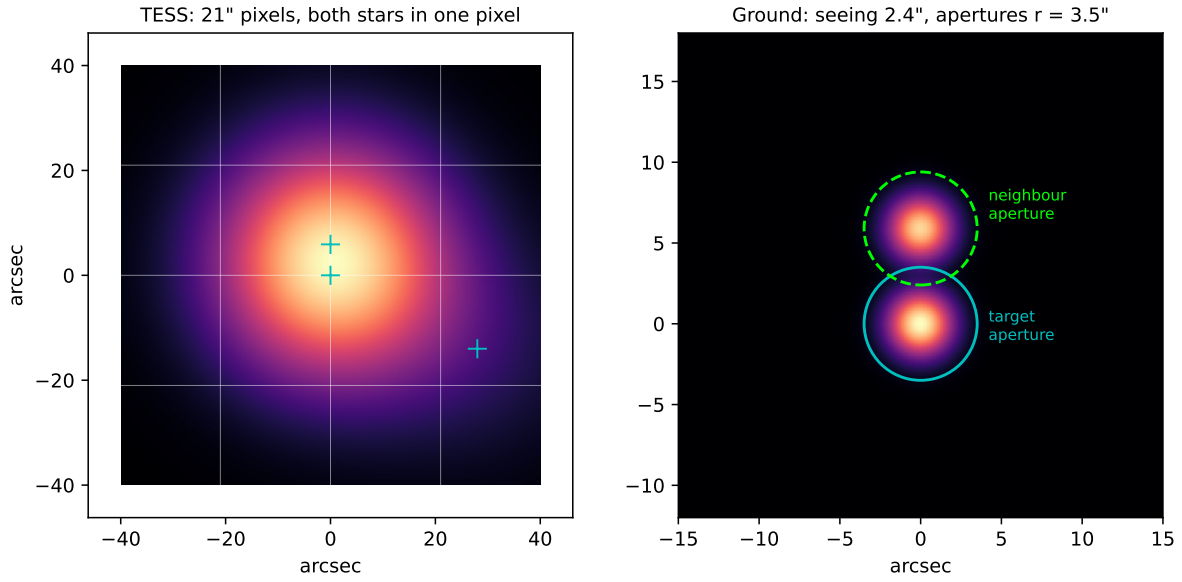


Figure 4.1: Simulated view of the field of TIC 65344135. Left: TESS, 21″ pixel grid; target and neighbour (crosses) fall in one pixel. Right: from the ground with 2.4″ seeing, with apertures of 3.5″ radius on the target and on the neighbour.

4.2 What the pipeline does with a ground light curve

1. **Fit.** The shape of the transit (duration, ingress and egress) is taken from the TESS fit. Only the depth and the mid-time are free, together with a baseline (linear in time, plus airmass

when present). The mid-time is searched within the uncertainty of the TESS ephemeris (at least ±60 min). The errors are inflated for correlated noise. Why the TESS shape is used is explained in the box below.

2. **Flux fractions.** For every star near the target, the fraction ϕ_j of the light in the ground aperture is computed. This uses the aperture radius, the seeing and the magnitude of each star in the observed band.

3. **Expected depth.** If star j hosts the signal, the ground aperture must show the depth $\delta_j = (\text{intrinsic depth required on } j) \times \phi_j$.

4. **Clearance.** Star j is *excluded* when this expected dip is absent (δ_j well above the 3σ upper limit), or when a dip is seen that is much deeper than δ_j. Excluded neighbours are removed before TRICERATOPS runs. The target is never removed, but if the ground data contradict it, the log warns loudly.

5. **Colour.** The intrinsic depth is compared between bands and with TESS. A planet's depth is almost the same in every band; an EB or blend often is not.

6. **Timing.** Good mid-times are added to the ephemeris and ETV analysis.

Tip. Why fit a ground light curve with the TESS shape?

- *The question is whether the TESS event is there.* The duration and the ingress/egress times are set by the geometry of the orbit. They are the same in every band and in every aperture; only the depth changes, with the dilution and slightly with limb darkening. A dip of the right star must therefore have the TESS shape.
- *TESS knows the shape much better.* TESS averages 7 transits. A single ground light curve has a few hours of data with trends from airmass and seeing, often only one side of baseline, and an S/N of 5–15. A free fit of duration and impact parameter to such data is badly degenerate with the baseline. Fixing the shape leaves two well-determined numbers, the depth and the mid-time.
- *Depths must be comparable.* The clearance compares the ground depth with the depth expected from TESS. That only works if both are measured with the same template.
- *Non-detections need a template.* "No dip" becomes a number, the 3σ upper limit, only for a dip of a given shape and duration.

Limitations. Limb darkening is stronger in blue bands, so the bottom of the transit is slightly more rounded there. This changes the fitted depth by a few per cent, which the 25% tolerance and the colour test absorb. A dip with a clearly different duration would fit the template badly (χ^2, residuals in `ground_lightcurves_<TIC>.pdf`). It would then not be the TESS signal, or the TESS shape would be wrong. Either way, look at the figure.

How to read the clearance diagram

The diagram answers one question per light curve and per star: *if this star were the source of the TESS dip, what would this ground aperture have to show?*

1. **The expected dip of a star.** TESS sees 3.2 ppt in the blended light. For each star, TRICERATOPS computes how deep its *own* eclipse would have to be to produce that: about 5.9 ppt for the target and 6.9 ppt for the neighbour. In a ground aperture, a star contributes only a fraction ϕ of the light (printed under its name). So the aperture would show $\phi\times$ that depth. This is the height of the point. For example, in the target aperture of the top panel the target supplies 99.4% of the light: $0.994 \times 5.9 \approx 5866$ ppm. The neighbour supplies 0.6%: 0.006×6.9 ppt ≈ 40 ppm.

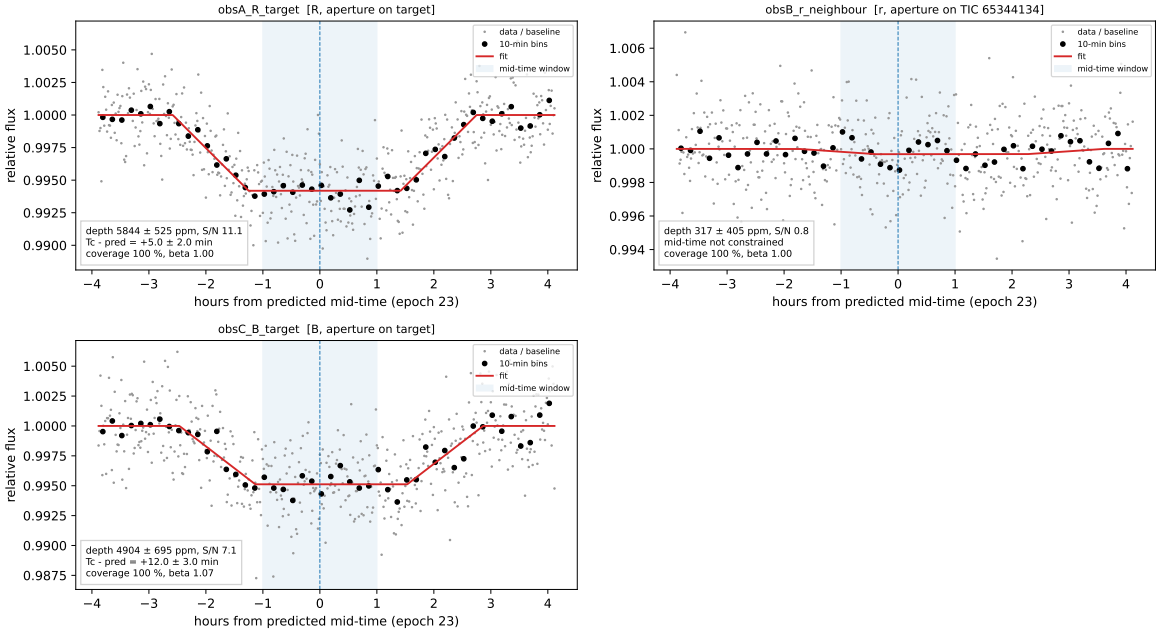


Figure 4.2: Ground light curves as fitted by the pipeline (synthetic test for TIC 65344135): R on the target (dip detected), r' on the neighbour (no dip), B on the target.

2. **What was measured.** The title of each panel says whether a dip was detected. If it was, the orange line and band show its depth and 3σ range. The dotted line is the largest dip still compatible with the data, anywhere in the allowed mid-time window.

3. **The verdict per star:**

   Consistent (green). The point lies in the orange band: the dip this star would cause is the dip that is seen. Top and bottom panels: the target.

   Excluded, "its dip would be seen, but is not". The point lies well above the dotted line: this star would have produced an obvious dip in its own aperture, and there is none. Middle panel: in the neighbour's own aperture it would cause 6943 ppm, but at most 1937 ppm is allowed. The neighbour cannot be the source.

   Excluded, "too faint here to cause the dip seen". A dip *is* detected, but this star supplies so little light to the aperture that, even as the host, it could cause only a tiny fraction of it. Top panel: the neighbour could cause 40 ppm in the target aperture, while 5844 ppm is seen.

   Allowed (blue). No conflict, but also no test: no dip is seen and the star is too faint in this aperture for its dip to be visible. Middle panel: the target, which supplies only 0.8% of the light in the neighbour's aperture.

4. **Combining the panels.** A star excluded in at least one light curve, and consistent in none, is removed before TRICERATOPS. Here the neighbour is excluded twice (top and middle panels) and the target is consistent twice, in R and in B. The target stays, and the neighbour's NTP/NEB scenarios disappear. If a star were excluded in one light curve and consistent in another, the pipeline would report a *conflict* and keep the star.

A tolerance of 25% on the expected depth (`--ground-depth-tol`) absorbs the uncertainties of the seeing model, the band magnitudes and limb darkening. A star is never excluded on a difference smaller than that.

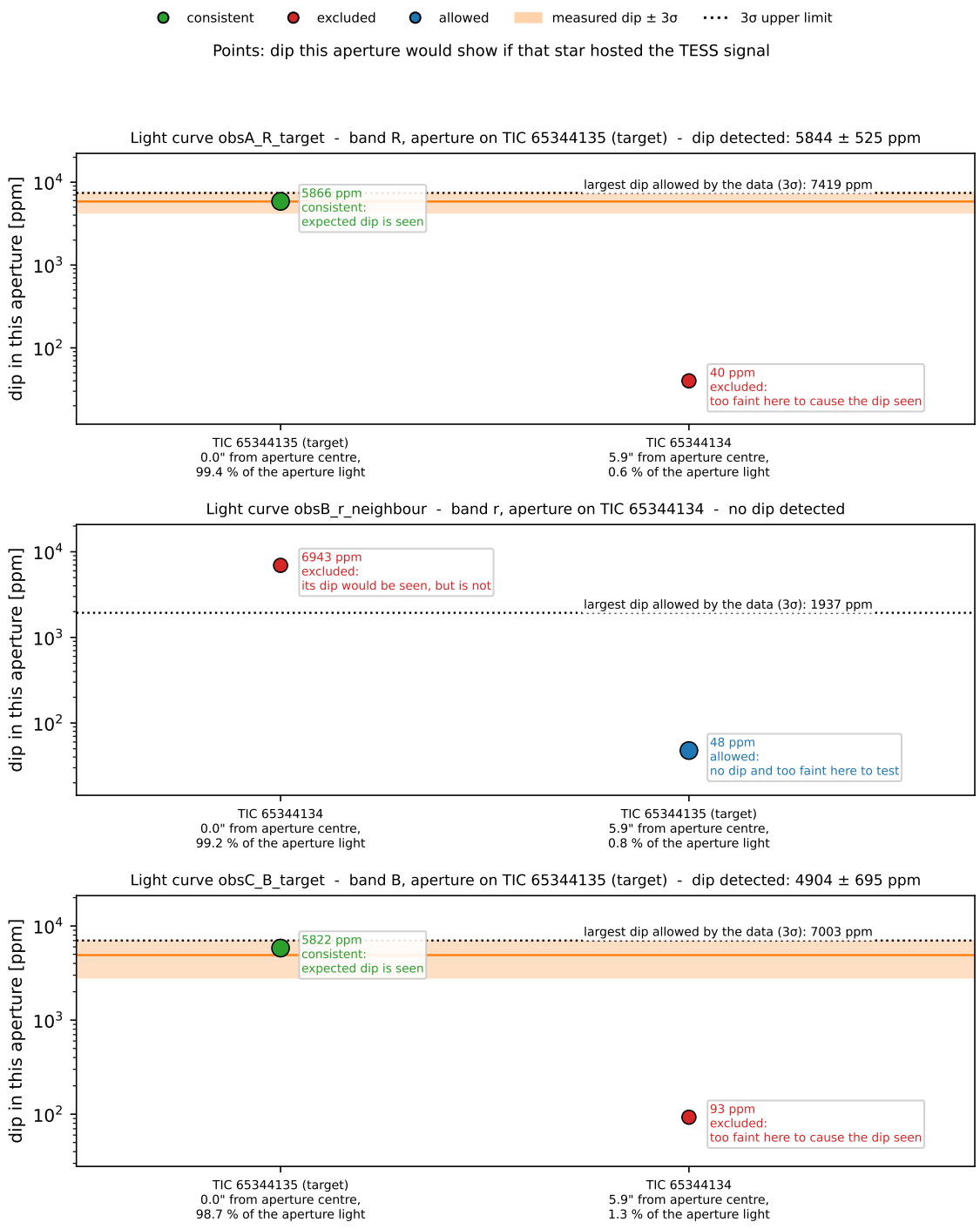


Figure 4.3: Clearance diagram (`ground_clearance_<TIC>.pdf`) for the three synthetic light curves of Figure 4.2. One panel per ground light curve. Each point is a star that could host the TESS signal. Its height is the dip *this* aperture would show if that star were the host. The orange band is the measured dip $\pm 3\sigma$ (only when a dip is detected); the dotted line is the largest dip the data allow. Colours: green *consistent*, red *excluded*, blue *allowed*. The reasoning is explained in the text.

4.3 Practical advice for observers

- **Timing.** Observe the *whole* transit with at least 45–60 min of baseline before ingress and after egress. Add the ephemeris uncertainty (in the log, or ask) to the predicted times. The GUI tool *Predicted transit times* gives the next events (Section 7).
- **Band.** A red band (Rc or Sloan r') gives the best S/N for most stars. A second, blue band (B, V, g') on another telescope or night allows the colour test.
- **Exposure.** Do not saturate. Cadences of 1–3 min are fine. The transit of TIC 65344135 lasts 5.3 h, so binning loses nothing.
- **Apertures.** Measure the target and *each* neighbour that could host the signal in separate apertures; in AstroImageJ, as targets T1, T2, ... Choose a radius of about 1.5–2 times the seeing FWHM but smaller than the separation, if possible. Report the radius and the FWHM in arcseconds.
- **Comparison stars.** Use several, of similar brightness and colour; reject variables.
- **Deliver raw relative fluxes.** Deliver them with an airmass column, not a light curve you have already detrended: the pipeline fits the baseline together with the transit.
- **Times.** Use BJD_TDB if possible. JD_UTC is accepted and converted, but say which one it is.
- **Formats.** AstroImageJ measurement tables, AAVSO/EXOTIC exoplanet reports or a simple table of time, flux and error are all read.
- **Precision.** In the synthetic test, 2 ppt per 1-min point gave the depth to ± 0.5 ppt. That is enough to tell 5.9 ppt from zero at more than 10σ, and 5.9 from 7.0 ppt marginally. Several light curves combine.

Part II

Using the GUI

Chapter 5

Installation and start

5.1 What you need

The scripts. `triceratops_validation.py`, `triceratops_gui.py`, the folders `profiles/` and `tests/`, all in one folder, here `D:\\EXPLORE\\candidates_dymock` (in WSL: `/mnt/d/EXPLORE/candidates_dymock`).

A Python environment for the pipeline. It needs triceratops 1.1, lightkurve, astroquery, wotan, transitleastsquares, numpy, scipy and matplotlib. Optionally, a second environment `triplus` holds TRICERATOPS+ (technical manual, Chapter 52):

```
conda create -n triplus -c conda-forge python=3.11 "numpy<2.4" scipy pandas matplotlib
    astropy astroquery numba celerite emcee seaborn arviz corner pytest
conda activate triplus
pip install lightkurve "pytransit==2.2.0" mechanicalsoup pyrr wotan transitleastsquares "
    setuptools<81"
git clone https://github.com/JGB276/TRICERATOPS-plus.git
cd TRICERATOPS-plus && pip install .
```

Tkinter for the GUI. Any conda Python has it. A system Python on Ubuntu/WSL needs `sudo apt install python3-tk`. On Windows 11 the GUI window of WSL appears on the desktop automatically (WSLg).

5.2 Starting

```
cd /mnt/d/EXPLORE/candidates_dymock
python triceratops_gui.py
```

The GUI reads the options from the validation script in the same folder and restores your last session. Check the installation once with *Tools* → *Run regression tests*; all tests should pass.

Chapter 6

A tour of the window

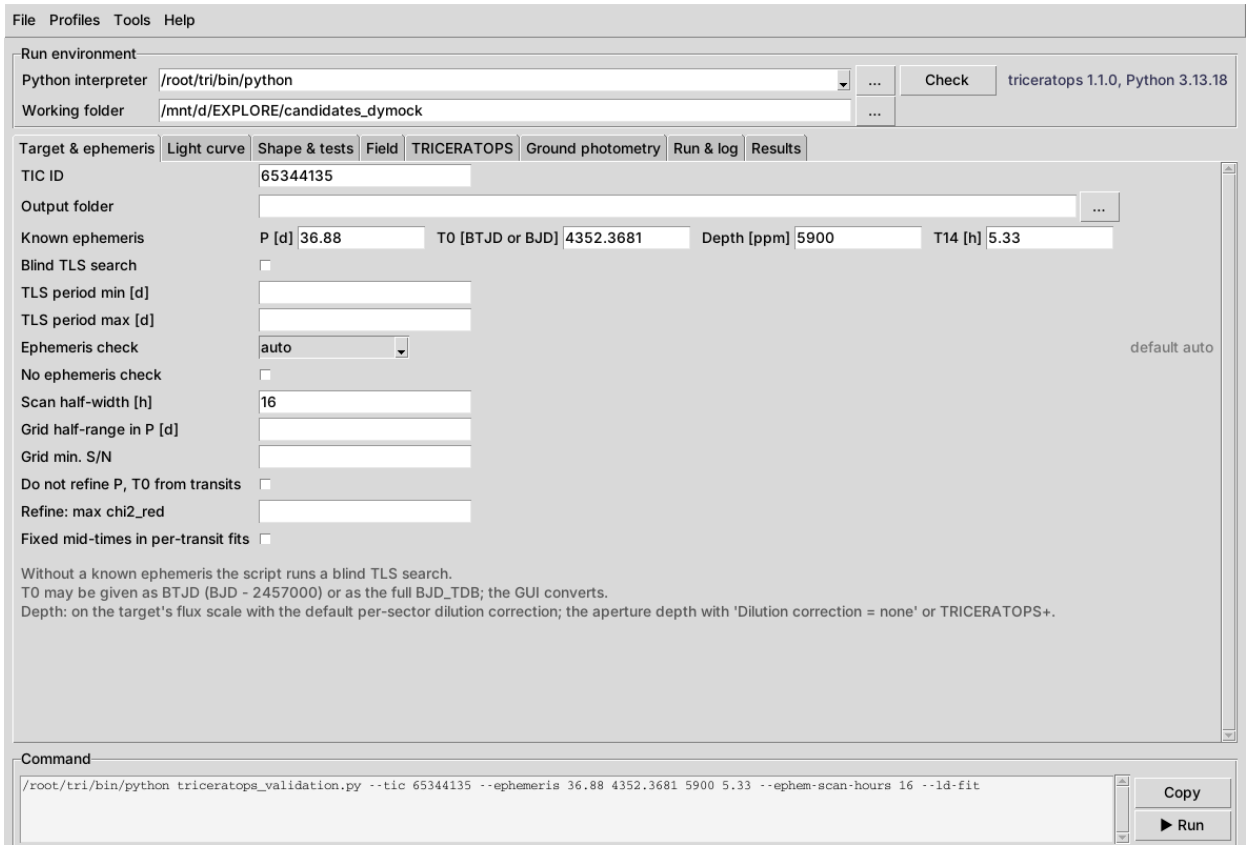


Figure 6.1: The main window with the profile of TIC 65344135 loaded.

From top to bottom (Figure 6.1):

Menu bar. *File* (profiles, import or copy a command), *Profiles* (ready-made settings per object), *Tools* (interpreter check, transit times, tests), *Help* (this manual and the technical one, command-line help).

Run environment. The *Python interpreter* that runs the pipeline. Choose your triceratops 1.1 environment, or `triplus` for TRICERATOPS+. *Check* shows which triceratops it contains. The *Working folder* is where the run starts; results go to `TIC<tic>/` in this folder unless an output folder is set.

Tabs. Six tabs with options, then *Run & log* and *Results*. Hover over an option's name to see its

command-line flag and explanation. Grey text on the right shows the default. Empty fields mean "use the default".

Command. The exact command that will run, updated as you type. Red text below it explains what is missing or inconsistent. *Copy* puts it on the clipboard, for a terminal or for your notes.

6.1 Profiles

A *profile* holds all the settings for one object. The menu *Profiles* (Figure 6.2) lists the profiles in the folder `profiles/`. Choosing one replaces all options: options not in the profile return to their defaults, so the same profile always gives the same command. The supplied profiles:

Profile	**Object and settings**
`profile_TIC65344135`	AAVSO campaign 942, triceratops 1.1: P = 36.88 d, T0 = BTJD 4352.3681, 5900 ppm (on the target's scale), T14 = 5.33 h, scan ±16 h, LD fit.
`profile_TIC65344135_triplus`	The same for the `triplus` environment (TRICERATOPS+): 3200 ppm (aperture scale), dilution correction none, TRICERATOPS+ on.
`profile_TIC229788204`	Grazing EB candidate, P = 93.83471 d; LD fit, sensitivity runs.
`profile_TIC165501544`	P = 132.42 d, 3228 ppm; grid search.
`profile_TIC229675482`	P = 145.841 d, 863 ppm; grid search.
`profile_TIC229701403_b`, `_c`	Two long-period candidates, separate output folders.
`profile_template_new_candidate`	Start for a new object.

Save your own with *File* → *Save profile*, which also stores the interpreter and working folder, or *Save target profile (options only)*, the version to pass on to colleagues. You are asked for a short description, shown in the menu.

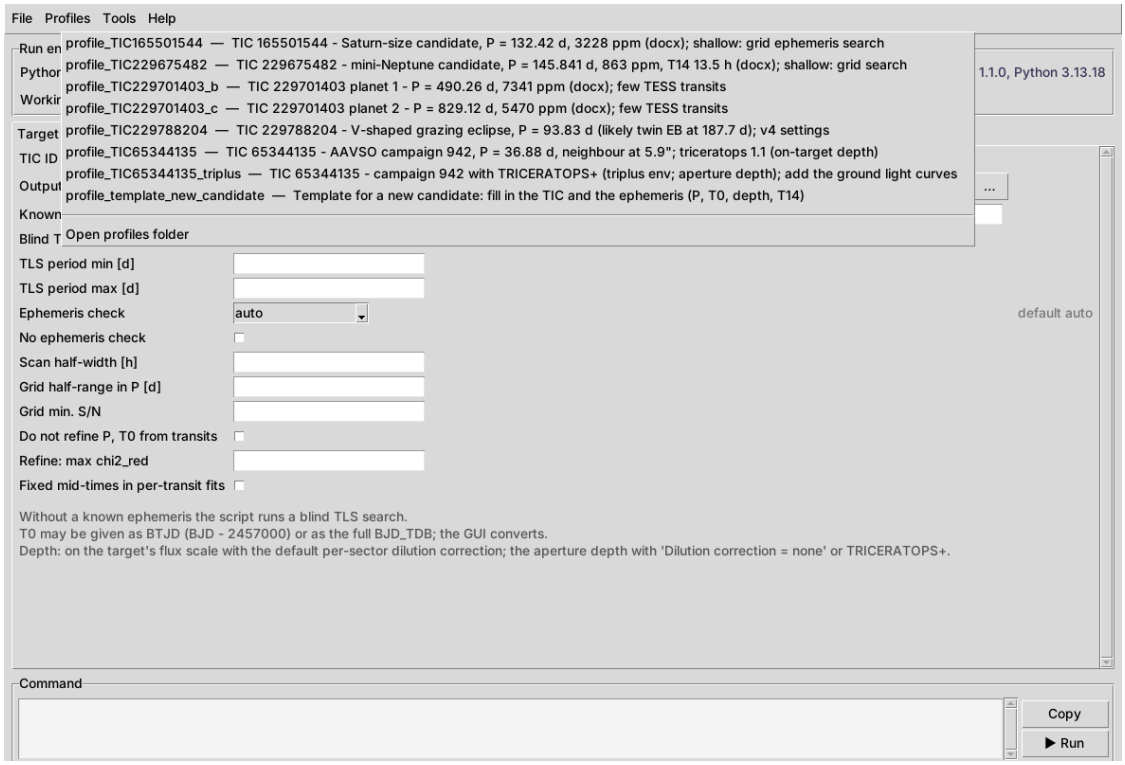


Figure 6.2: The *Profiles* menu.

Chapter 7

Worked example: AAVSO campaign 942 (TIC 65344135)

Situation. TESS shows six transits of 3.2 ppt with P = 36.88 d. The neighbour at 5.9″ could be the source. Dennis Conti's campaign asks for ground photometry of the transits of 7 November and 14 December 2026, with target and neighbour measured separately, in a red and if possible a blue band.

Step 1 — Load the profile and check the environment

1. *Profiles* → *profile_TIC65344135*. For TRICERATOPS+ choose `profile_TIC65344135_triplus` instead.
2. Choose the matching interpreter and press *Check*. The text next to it reads, for example, "triceratops 1.1.0, Python 3.13" or "TRICERATOPS+ 1.0.17, Python 3.11".
3. Look at the tab *Target & ephemeris* (Figure 6.1). The ephemeris fields hold P, T0, depth and T14. T0 may be typed as the full BJD from the campaign page (2461352.3681); the GUI converts it to BTJD and says so under the command.

Step 2 — Plan the observation (optional)

Tools → *Predicted transit times* lists the next transits from the ephemeris fields (Figure 7.1). The UTC times are approximate (±8 min, no barycentric correction); for 7 November the tool gives 20:48 UTC, the campaign page 20:43 UTC. Add the ephemeris uncertainty and the baseline before and after.

Step 3 — Add the ground light curves

Put the observers' files in a subfolder, for example `campaign942/`. On the tab *Ground photometry* (Figure 7.2) press *Add files*, select one or more files and fill in the dialog for each (Figure 7.3):

Field	What to enter
Band	R (= Rc), V, B, I, or Sloan g r i z; also J H K, CBB, clear. Empty: from the file header, else R.

Field	What to enter
Aperture star (TIC)	Empty for the target; 65344134 for the neighbour's aperture.
Aperture radius	Radius in arcsec, as used in the photometry.
Seeing FWHM	In arcsec (median of the night).
Observer	Name or AAVSO code.
Time system	Empty or BJD_TDB; JD_UTC or HJD_UTC if that is what the file has.
Flux / error / time column	Only if the defaults are not right, e.g. `rel_flux_T2` for the neighbour in an AstroImageJ table where it is T2.
Magnitudes	1 if the flux column holds magnitudes.
Detrend	Empty (= linear + airmass); `poly2` or extra columns, e.g. `poly1, AIRMASS,FWHM_Mean`.
Centre RA/Dec, pixel scale	Rarely needed (aperture not on a catalogued star; AstroImageJ radii in pixels).
TRICERATOPS+ band	Only to override the automatic choice.

The same AstroImageJ file can be added twice, once for the target (T1) and once for the neighbour (`fcol=rel_flux_T2`, aperture star 65344134). *Save as list* writes the table to a list file that can be used on the command line (`--ground-list`); *Import list* reads one.

Step 4 — Save, run and follow

1. *File* → *Save profile* keeps this set-up, for example as `profile_TIC65344135_campaign.json`.
2. Press *Run*. The tab *Run & log* shows the output as it comes (Figure 7.4): section headers in blue, warnings in red. The status line shows the elapsed time and the current step.
3. Typical durations:
   - the first download of all TESScut sectors: tens of minutes; later runs use the cache;
   - the light-curve analysis: a few minutes;
   - the TRICERATOPS runs: up to the time budget, 1 h by default.
4. *Stop* ends a run. You can close the window only after confirming.

Step 5 — Read the results

When the run ends, the tab *Results* opens (Figure 7.5). On the left is the summary as a tree, on the right the output files; a double click opens one. The key items:

Summary item	How to read it
`ld_fit`	Radius ratio, impact parameter b, `grazing`. A grazing fit ($b+k > 1$) favours an EB.
`secondary`	Best secondary eclipse and the 3σ depth limit; a detection means EB.
`etv`	O−C scatter, odd/even timing, limits on a third body.

Summary item	How to read it
`ground` → `target_status`	*consistent*: the target shows the expected dip. *excluded*: it does not, so a likely false positive (check the data first!).
`ground` → `cleared`	Neighbours excluded by the ground data and removed before TRICERATOPS.
`ground` → `chromaticity`	p of a band-independent depth; `chromatic = True` means EB or blend.
FPP, NFPP, verdict	See Chapter 2: PASSED when FPP < 0.015 and NFPP < 0.001; "not converged" means provisional.
`scenario_classes`	Probability per family: planet on the target, EB on the target/-companion, background star, nearby star.

How to interpret the outcomes for TIC 65344135

Ground result	Consequence
Target: dip of ≈5.9 ppt; neighbour: flat	Neighbour cleared, NFPP ≈ 0. The FPP is then decided by the shape of the transit and the remaining scenarios on the target.
Target flat; neighbour: dip of ≈7 ppt	Nearby eclipsing binary: false positive. The log says "probable nearby eclipsing binary".
Dip on the target but much shallower or deeper than 5.9 ppt	Check the aperture radius, the comparison stars and the time window. If the result is confirmed, the dilution model or the host is wrong.
Depth differs between red and blue	Chromatic: EB or blend, even if the dip is on the target.
Light curves do not cover the transit	*unconstrained*: no conclusion; a later transit is needed.

Example from a real run. For TIC 229788204 (TESS only) the pipeline found a V-shaped, grazing eclipse with no secondary, an FPP of 1.00, and probabilities of 0.55 for EB, 0.31 for EBx2P and 0.14 for PEBx2P. This is the typical signature of an eclipsing binary, here most likely a pair of nearly equal stars at twice the period (187.7 d).

Note. The numbers in Figures 4.2, 4.3 and 7.5 come from a *synthetic* test that mimics the campaign. The first real TESS-only result for this star is in Chapter 8.

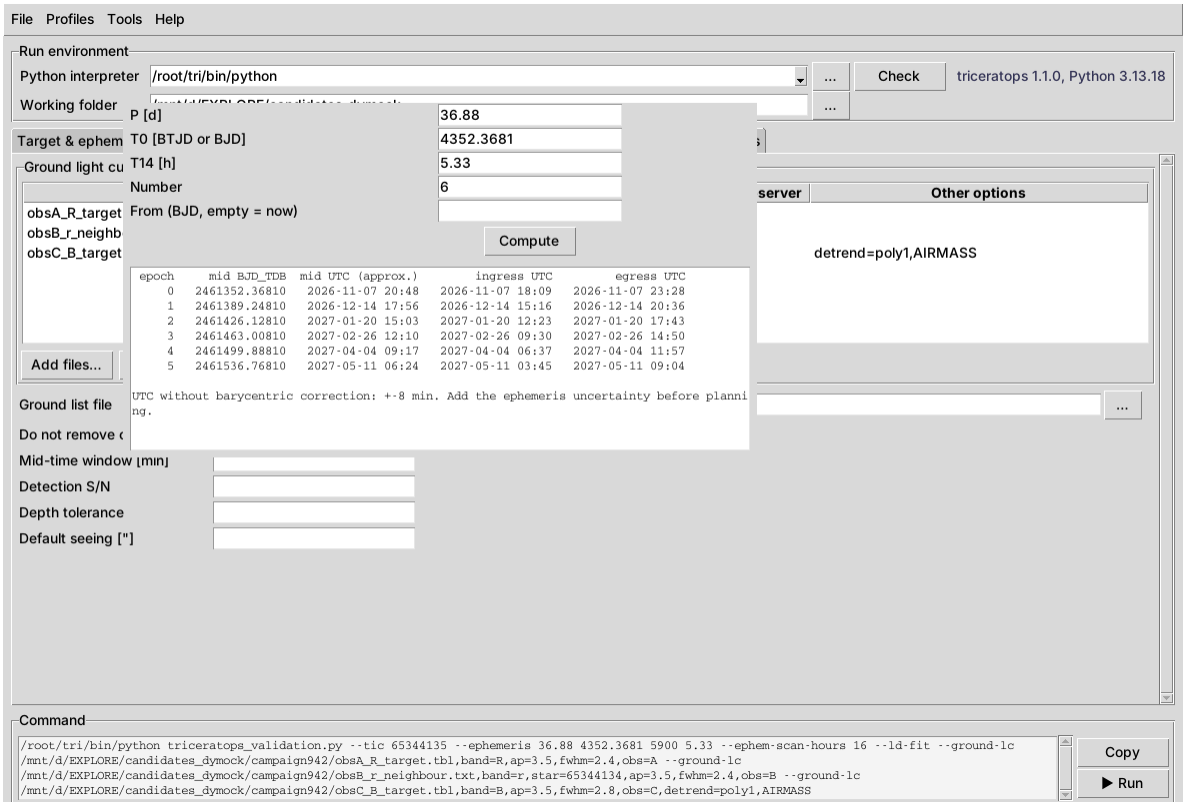


Figure 7.1: Predicted transits of TIC 65344135.

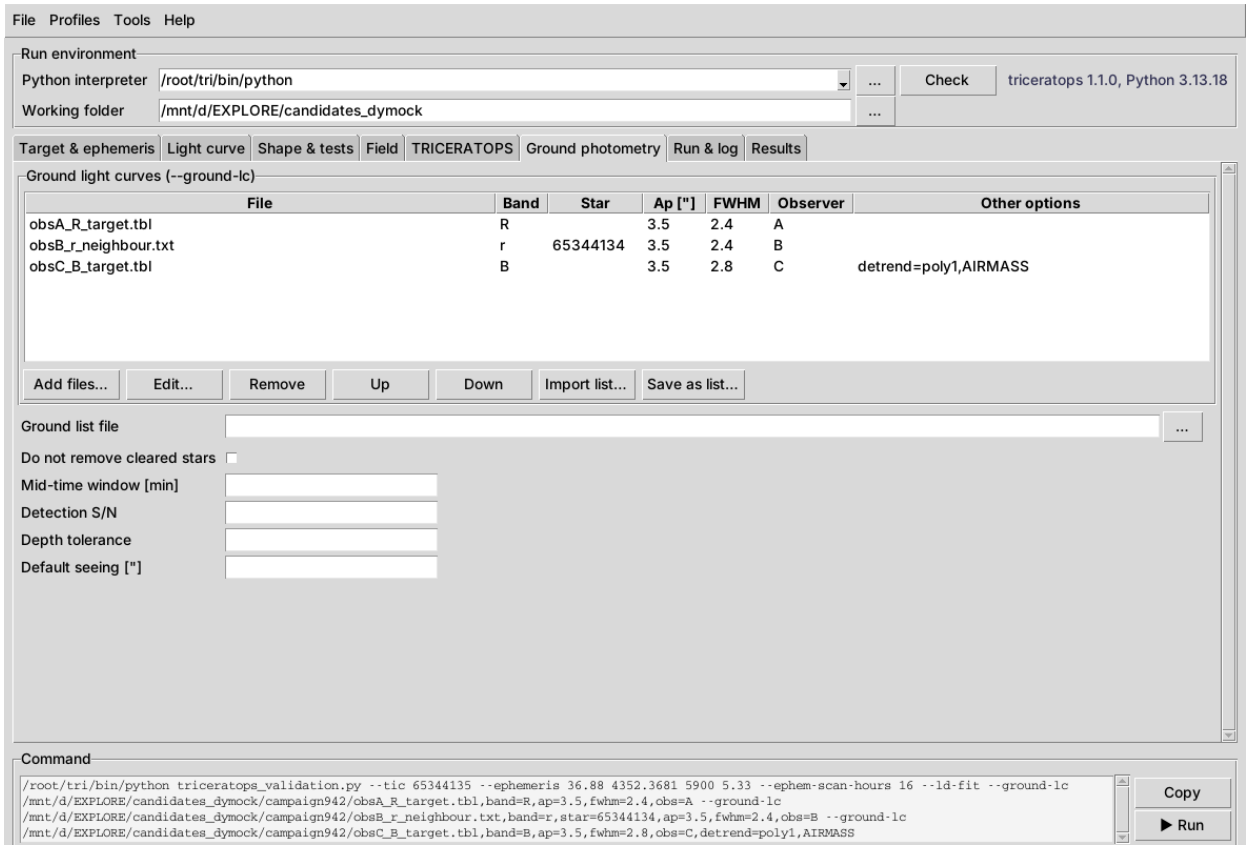


Figure 7.2: Tab *Ground photometry* with three light curves.

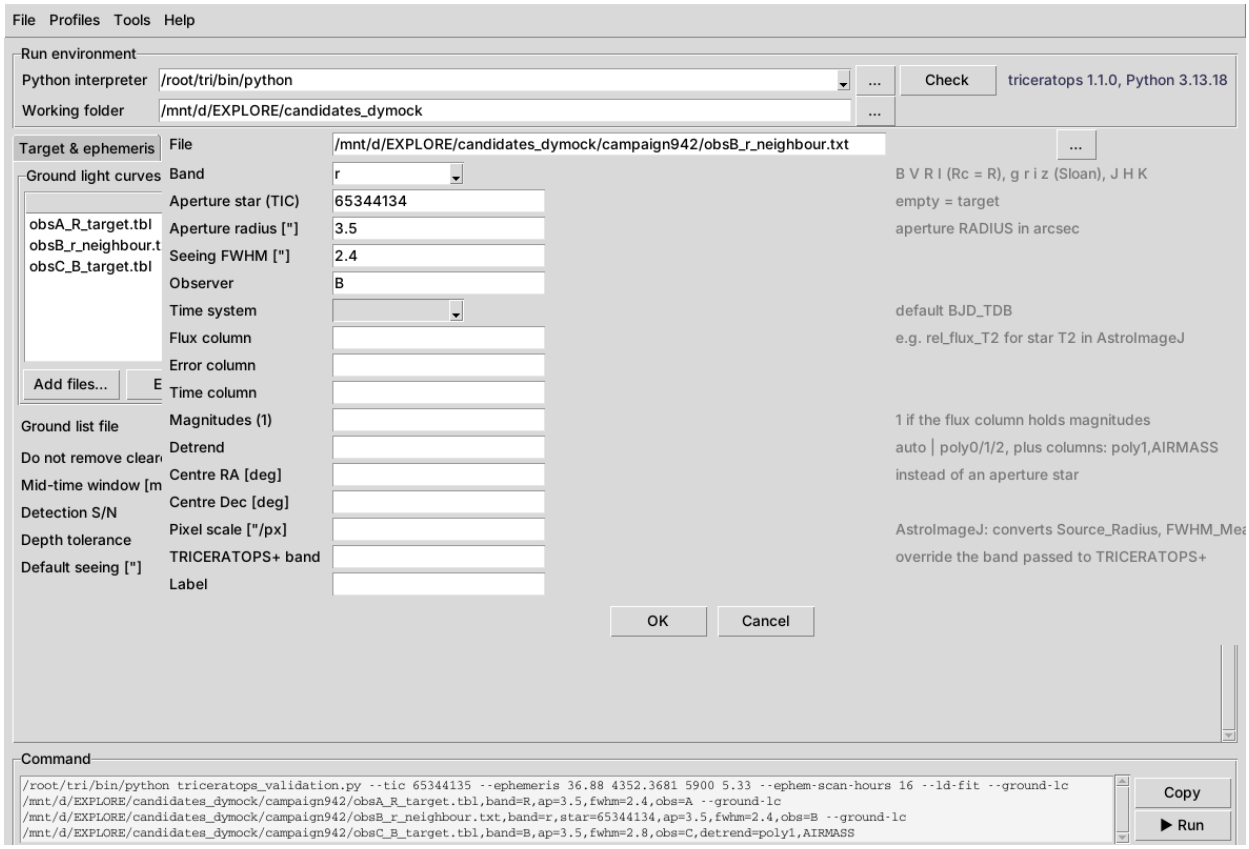


Figure 7.3: The dialog for one ground light curve (the neighbour's aperture).

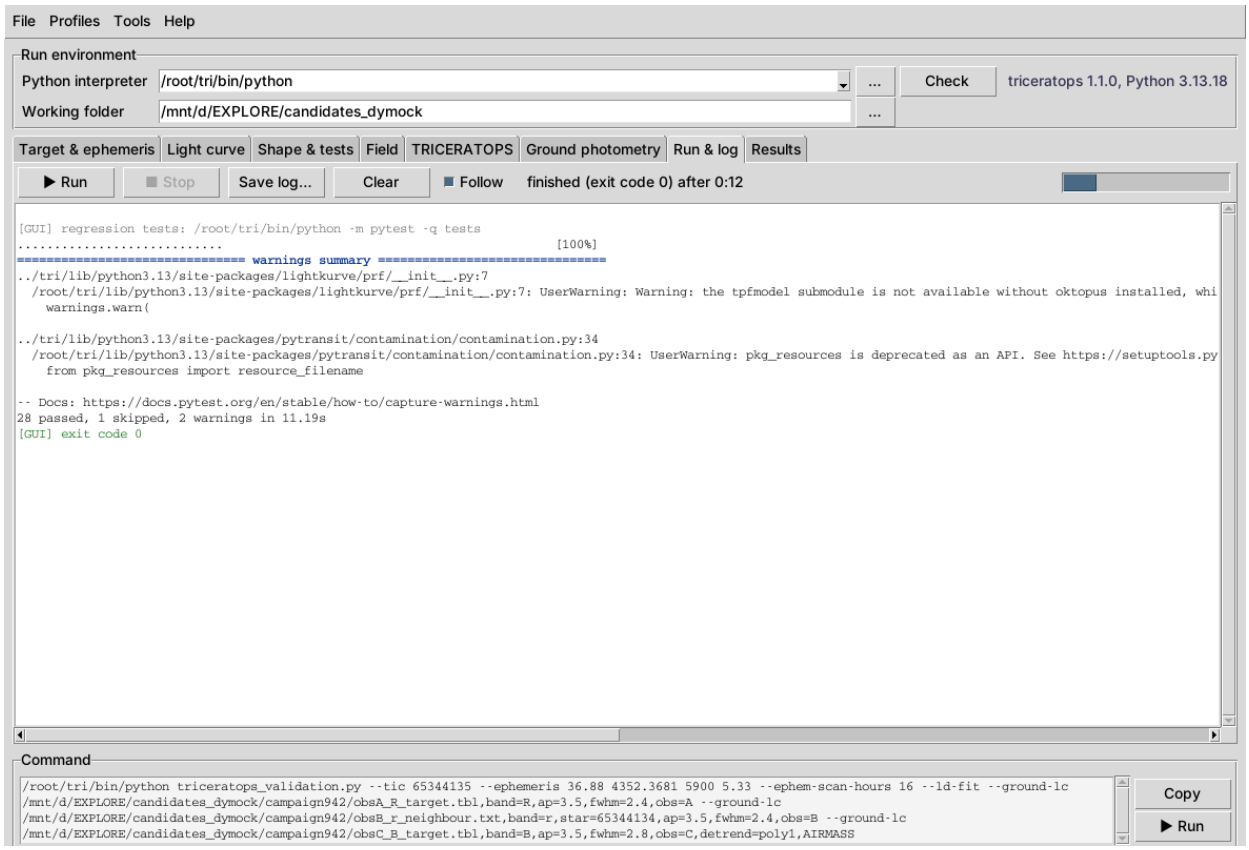


Figure 7.4: Tab *Run & log* (here after running the regression tests).

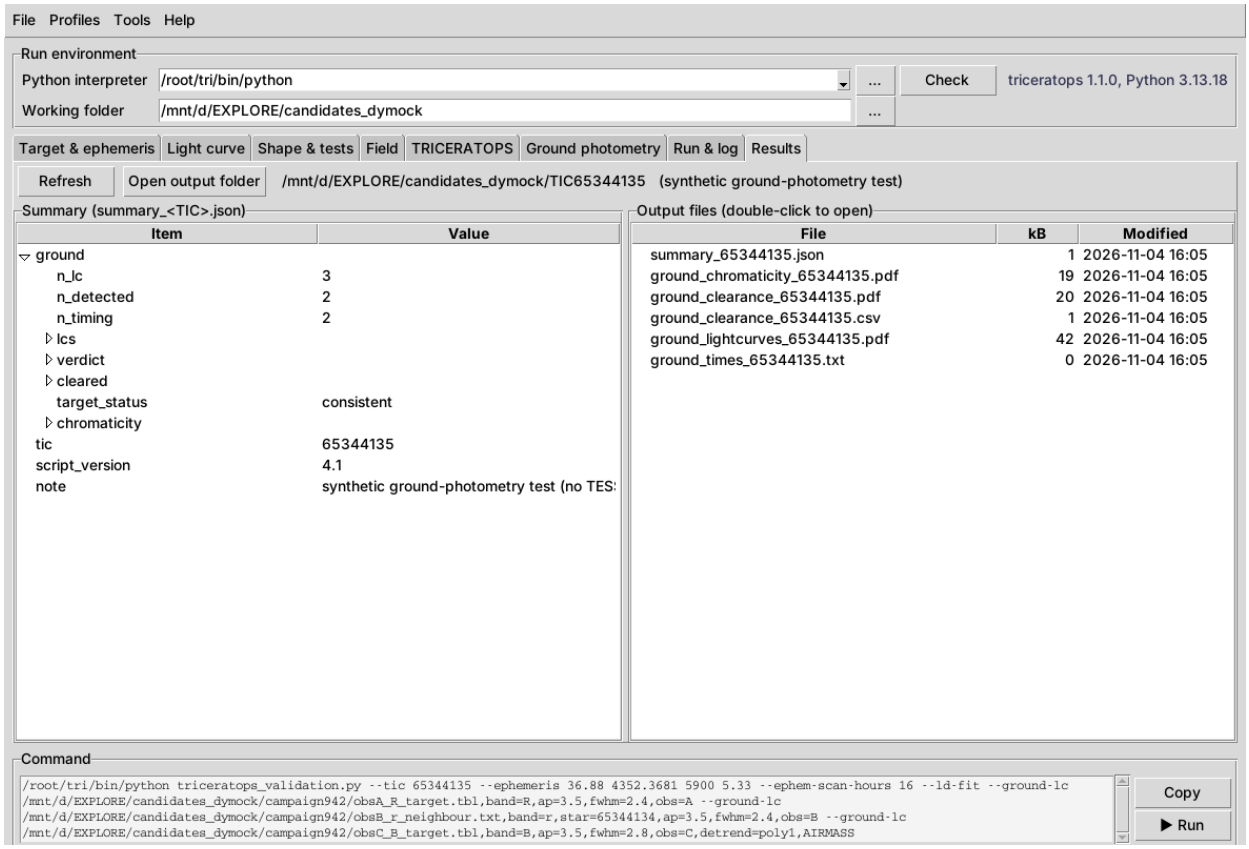


Figure 7.5: Tab *Results*: summary and output files (here of the synthetic ground-photometry test).

Chapter 8

Preliminary result: TIC 65344135 with TESS only

This chapter reports the first complete run on the real TESS data of TIC 65344135, made on 4 November 2026 before the campaign transits. It is the baseline that the ground-based light curves will improve on. The figures are the pipeline's own output for this star.

Set-up

- **Command.** Profile `profile_TIC65344135` in the `triplus` environment (TRICERATOPS+ 1.0.17, NumPy 2.3, pytransit 2.2):

```
triceratops_validation.py --tic 65344135 --ephemeris 36.88 4352.3681 5900 5.33 --ephem-scan-hours 16 --ld-fit --tri-plus on
```

- **Data.** TESScut full-frame images of 7 sectors (3, 4, 30, 97, 106, 107, 108; 2018–2026), with cadences of 30, 10 and 3.3 min.
- **Dilution.** In every sector the optimal aperture contains the target and the neighbour TIC 65344134 (5.9″). The target supplies a fraction $g = 0.556$–0.562 of the light in the aperture (Figure 8.1).
- **TRICERATOPS.** 20 runs, all valid, evidences pooled (re-run of 4 November with the corrected script); no ground-based light curves yet.

Results

Quantity	Result
Transits in TESS	7 (epoch −2, sector 107, weak: S/N 2.6).
Refined ephemeris	P = 36.87868 ± 0.00006 d, T0 = BJD 2461352.37156 ± 0.00225 (3.2 min).
Depth in the TESS aperture	3060 ppm (limb-darkened fit); 2825 ± 117 ppm (per-transit mean).
Depth on the target alone	5320 ± 210 ppm if the target is the host (aperture depth / g).
Duration T14	5.49 h.

Quantity	Result
Shape	U-shaped, not grazing; $b \approx 0$ (poorly constrained). $R_p/R_\star = 0.051$ on the aperture scale, 0.068 after the dilution correction ($/\sqrt{g}$). With the TIC radius of 0.99 $R_\odot$ that is about 7.3 $R_\oplus$ if the target is the host (TRICERATOPS, TP scenario: 7.5 $R_\oplus$), a sub-Saturn.
Odd/even depths	0.9σ difference: no sign of an EB at twice the period.
Secondary eclipse	None; 3σ limit 450 ppm. An eclipsing companion would be cooler than about 3560 K.
Timing (ETV)	O−C rms 9.4 min; no significant period change or third body.
Difference imaging	Centroid consistent with the target (median 0.9σ), but $5.9''$ cannot be resolved this way.
FPP (pooled over 20 runs)	0.389 ± 0.014 (run-to-run scatter 0.06).
NFPP (pooled over 20 runs)	0.376 ± 0.014.
Verdict	NOT passed (requires FPP < 0.015 and NFPP < 0.001).

What the probabilities say

The probability is shared between two explanations (Figure 8.4):

- a planet on the target (TP, PTP, DTP): 0.61;
- a planet on the neighbour TIC 65344134 (NTP): 0.38.

The evidences of the TP and NTP scenarios are practically identical (pooled: 0.43 for TP, 0.38 for NTP). This is expected: the two stars are almost twins (Tmag 12.33 and 12.56, $T_{\rm eff}$ 5540 and 5390 K), so the same planet fits on either. Eclipsing-binary scenarios together have 1.4%, background stars 0.02%.

The light curve itself therefore looks like a planet; what TESS cannot tell is around which star. The FPP is high only because of the neighbour (FPP ≈ NFPP).

What the campaign can decide

Ground result	Expected effect
Neighbour flat, target dip ≈5.3 ppt	TIC 65344134 removed: NFPP → 0. FPP ≈ 0.014/0.625 ≈ 0.02: a *likely planet*, but not yet below 0.015.
Target flat, neighbour dip ≈6.9 ppt	Nearby eclipsing binary: the candidate is a false positive.
Depth differs between red and blue	Chromatic: blend or EB, even on the target.

After clearing the neighbour, the remaining 2% come mostly from scenarios with an unresolved companion of the target: STP (planet on the companion) and SEB. Two kinds of follow-up reduce them:

- a contrast curve from high-resolution imaging, which limits companions;

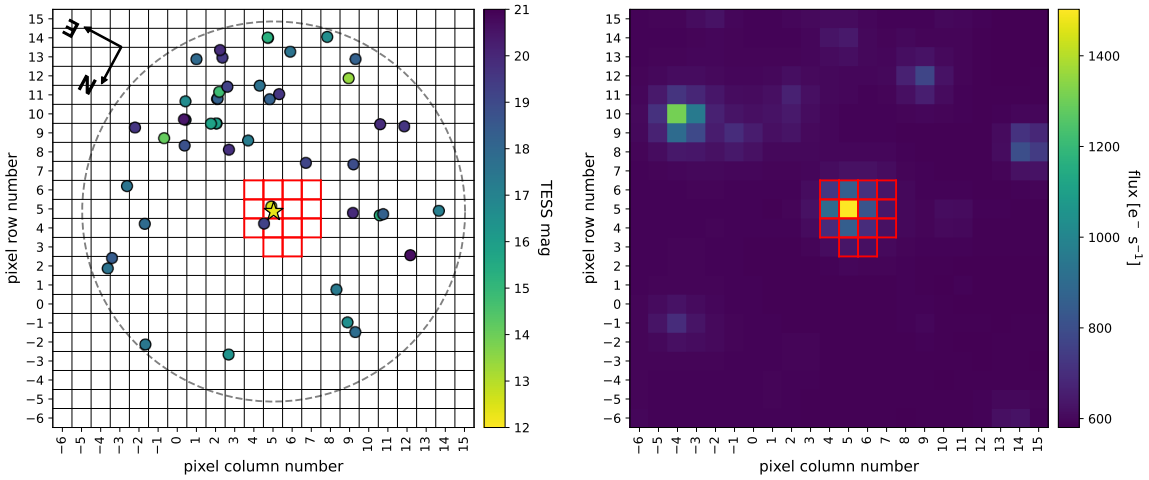


Figure 8.1: Sector 97: TIC stars on the TESS pixel grid (left) and the image with the optimal aperture (red, right). Target and neighbour are 0.28 pixel apart.

- multi-colour ground light curves in the TRICERATOPS+ likelihood.

With one of these, validation (FPP < 0.015) is realistic.

Updated times for the observers

The refined ephemeris puts the transits about 5 min later than the campaign page. UTC times are derived from the campaign page's conversion of BJD to UTC, with T14 = 5.49 h. Add ±3 min uncertainty plus at least 1 h of baseline before and after.

Date	Mid BJD_TDB	Ingress	Mid	Egress	vs. campaign
7 Nov 2026	2461352.3716	18:03 UTC	20:48 UTC	23:33 UTC	+5.0 min
14 Dec 2026	2461389.2502	15:11 UTC	17:56 UTC	20:40 UTC	+5.5 min

The ground aperture should show these depths:

- on the target alone, about 5.3 ppt if it is the host;
- on the neighbour alone, about 6.9 ppt if it is the host;
- in an aperture holding both stars, about 3.1 ppt.

Remarks

- **Pooled FPP.** In the first run of 4 November the pooled FPP (0.335, bootstrap error 0.000) was not valid: with TRICERATOPS+ the evidences of the first run were reused for all 20 runs. The numbers in this chapter come from the re-run with the corrected script. The pooled evidences are now consistent with the mean over the runs (0.39), and no single run dominates (largest share 0.07).
- **Stellar density.** The density from the transit shape (2.2 $\mathrm{g\,cm^{-3}}$) is somewhat higher than that from the TIC mass and radius (1.4 $\mathrm{g\,cm^{-3}}$). With b unconstrained and a possible eccentric orbit, this is no reason for concern.

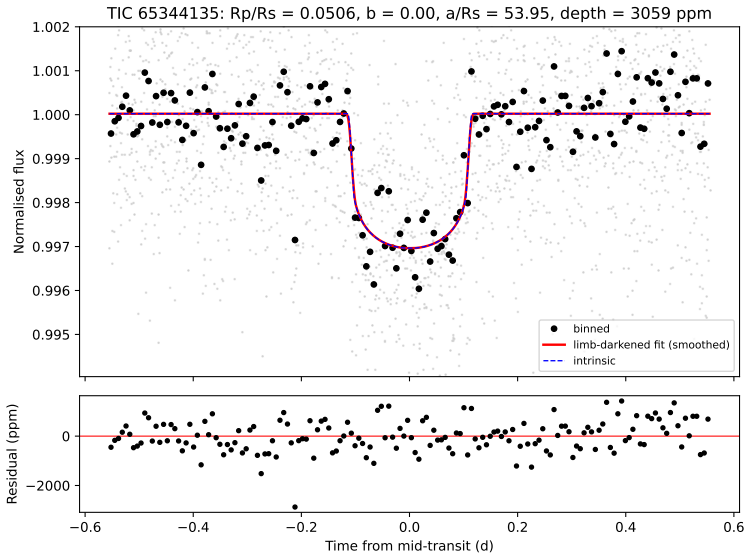


Figure 8.2: Folded TESS light curve of the 7 transits (aperture scale; 10-min bins) with the limb-darkened fit: a flat-bottomed transit of 3060 ppm and 5.5 h.

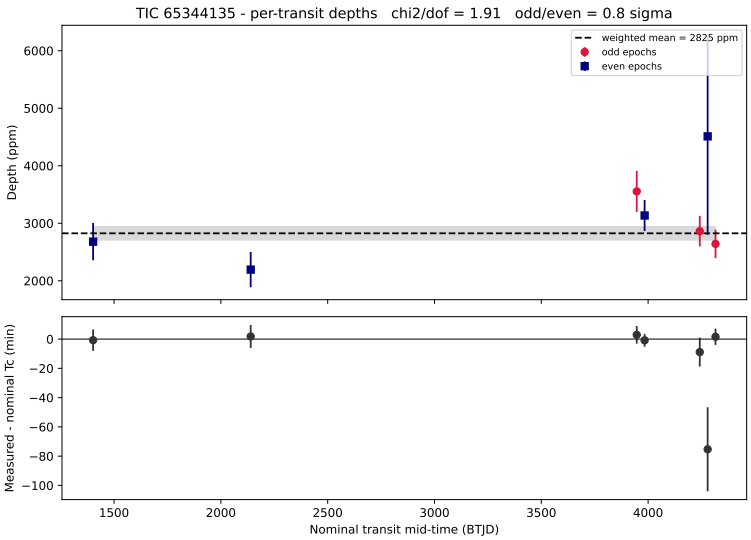


Figure 8.3: Depth (top) and mid-time offset (bottom) of the individual transits. Odd and even transits agree; the outlier at BTJD 4279 is the weak sector-107 event.

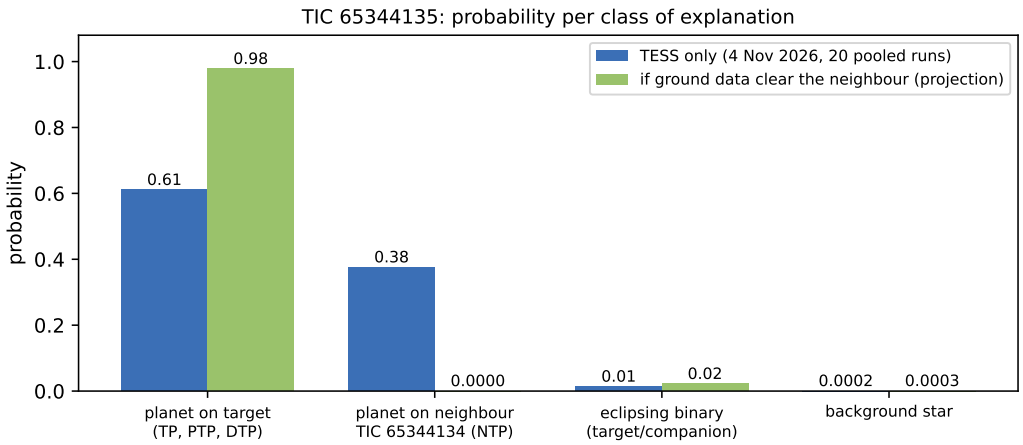


Figure 8.4: Probability per class of explanation from the TESS data (blue), and the projection if the ground photometry clears the neighbour (green).

Chapter 9

Other useful functions

Import a command line. *File* → *Import command line*: paste a command, from a colleague or from the technical manual, and the fields are filled in.

Copy command. To run the same analysis in a terminal or on another computer.

New. Resets all options.

Last session. Saved automatically in `~/.triceratops_gui.json`.

Regression tests. *Tools* → *Run regression tests* checks the installation in the selected interpreter (about 30 tests, ∼1 min).

Manuals. *Help* → *Observer's Guide (PDF)* opens this guide, *Help* → *Technical manual (PDF)* the full technical manual.

Part III

Reference

Chapter 10

All options

The tables list every option of `triceratops_validation.py` v4.1.1, grouped as on the tabs of the GUI. They were generated from the script itself. "–" means that the script chooses the value itself, as described in the explanation. In the GUI, leave such a field empty.

> **Tip. The options most observers need:** TIC ID, known ephemeris, scan half-width, limb-darkened fit, ground light curves, and for TRICERATOPS+ the dilution correction *none* with the aperture depth. Everything else can stay at its default.

Tab: Target & ephemeris

GUI field	Command line	Default	Meaning
TIC ID	`--tic TIC`	required	TIC ID of the target.
Output folder	`--outdir OUTDIR`	–	Output directory (default: TIC{tic}/ in the CWD). Created automatically if missing.
Known ephemeris	`--ephemeris P_D T0_BTJD DEPTH_PPM T14_H`	–	Known ephemeris for THIS target (skips the TLS search): period [d], T0 [BTJD], depth [ppm], duration [h].
Blind TLS search	`--blind-search`	off	Force a blind TLS search (default when `--ephemeris` is not given).
TLS period min [d]	`--p-min D`	–	TLS period_min [d] (default: 0.5). Narrow the grid when you know the approximate period.
TLS period max [d]	`--p-max D`	–	TLS period_max [d] (default: 100, also clipped to baseline/n_transits_min at runtime).
Ephemeris check	`--ephem-search auto|scan|grid|off`	auto	With `--ephemeris`: how to check/refine P and T0 before detrending. scan = box search of each transit + robust linear fit; grid = folded box search on a (P, T0) grid (stacks all transits, for shallow signals); auto = scan, grid if the scan is inconclusive (default); off = keep the input.
No ephemeris check	`--no-ephem-scan`	off	Same as `--ephem-search` off.
Scan half-width [h]	`--ephem-scan-hours H`	–	Half-width of the search in T0 [h], for scan and grid (default: max(2 x T14, 12 h)).
Grid half-range in P [d]	`--ephem-grid-dp D`	–	Grid search: half-range in P [d] (default: the T0 half-range divided by the number of epochs from the reference epoch).

GUI field	Command line	Default	Meaning
Grid min. S/N	`--ephem-grid-min-snr X`	–	Grid search: minimum stacked S/N to adopt the result (default 7).
Do not refine P, T0 from transits	`--no-refine-ephemeris`	off	Do not refine P and T0 from the individual transit mid-times.
Refine: max chi2_red	`--refine-chi2-max X`	–	Max chi2_red of the linear timing fit before the refined ephemeris is rejected (default: 8). Raise for long-period targets with few transits.
Fixed mid-times in per-transit fits	`--no-tc-fit`	off	Keep the mid-time of each individual transit fixed at the ephemeris value.

Tab: Light curve

GUI field	Command line	Default	Meaning
Light-curve source	`--lc-source tesscut|sap|pdcsap`	tesscut	Photometry: tesscut (FFI custom aperture, default), sap (SPOC SAP), or pdcsap (SPOC PDC). For sap/pdcsap the SPOC TPF is still fetched when available.
TESScut cache folder	`--download-dir DOWNLOAD_DIR`	–	TESScut / SPOC cache directory (default: <outdir>/tesscut_cache). Already-downloaded products are reused.
Exclude sectors	`--exclude-sectors S ...`	–	Sector numbers to drop before extraction (e.g. `--exclude-sectors` 56 79).
No automatic sector exclusion	`--no-auto-exclude-sectors`	off	Do not auto-drop sectors that fail normalisation quality cuts after detrending.
Max sector rms	`--sector-max-rms SECTOR_MAX_RMS`	–	Max robust RMS of norm_flux (fractional; default 0.02).
Aperture	`--aperture optimal|percentile`	optimal	TESScut aperture: S/N-optimal from a catalogue PSF model (default) or pixels above the 85th percentile.
Dilution correction	`--dilution-mode sector|none`	sector	sector (default): put every sector on the target's flux scale with its own target fraction before folding; none: keep raw aperture flux.
Raw aperture flux	`--raw-aperture-flux`	off	Pass dilution_corrected=False to triceratops (flux is an uncorrected aperture LC). Needs triceratops >= 1.1.
Light curve is dilution corrected	`--dilution-corrected`	off	Tell TRICERATOPS the light curve is crowding-corrected (automatic with `--dilution-mode` sector and for pdcsap).
Detrend windows [d]	`--detrend-windows D ...`	–	First-pass biweight windows [d]; the blind TLS search runs once per window and the highest-SDE candidate is kept (e.g. 0.75 1.0 1.5 2.5).
Error floor [ppm]	`--err-floor-ppm ERR_FLOOR_PPM`	0.0	Lower bound on the flux uncertainty given to calc_probs [ppm].
Do not inflate sigma	`--no-inflate`	off	Do not inflate sigma to chi2_red = 1 of the reference model.

Tab: Shape & tests

GUI field	Command line	Default	Meaning
Limb-darkened fit	`--ld-fit`	off	Fit the folded LC with a limb-darkened transit model (u1,u2 fixed to the Claret values of the TIC/TLS catalogue) and use it as reference model for the sigma inflation.
LD coefficients free	`--ld-free`	off	Like `--ld-fit`, but fit the limb-darkening coefficients (Kipping q1,q2).
LD coefficients u1 u2	`--ld-coeffs U1 U2`	–	Like `--ld-fit`, with fixed quadratic coefficients.
No secondary-eclipse search	`--no-secondary-search`	off	Skip the secondary-eclipse search at all phases.
Secondary min. S/N	`--secondary-min-snr X`	–	Stacked S/N for a secondary-eclipse detection (default 7).
Red-noise scrambles	`--red-noise-scrambles N`	0	Phase-randomisation trials for a red-noise false-alarm p-value on the stacked depth (0 disables; try 1000–5000).
Red-noise seed	`--red-noise-seed RED_NOISE_SEED`	42	RNG seed for `--red-noise-scrambles`.
No timing analysis	`--no-etv`	off	Skip the eclipse-timing / depth-variation analysis.
Extra mid-times file	`--extra-times FILE`	–	Ground-based mid-times to add to the timing analysis: 'BJD_TDB err_min [label]' per line.

Tab: Field

GUI field	Command line	Default	Meaning
Contrast curve	`--contrast-curve FILE`	–	High-resolution imaging contrast curve: separation [arcsec], delta mag.
Vet neighbours	`--vet-neighbours`	off	Vet nearby stars from their own TESS light curves and exclude those that cannot host the signal.
No difference imaging	`--no-difference-imaging`	off	Skip the difference-image centroid analysis.
Exclude stars (TIC)	`--exclude-stars TIC ...`	–	Stars cleared by other follow-up (removed before calc_probs).

Tab: TRICERATOPS

GUI field	Command line	Default	Meaning
Min. runs (n_iter)	`--n-iter N_ITER`	20	Minimum number of TRICERATOPS calc_probs runs (more are added until converged).
Samples per run (N)	`--n-samples N`	–	Monte-Carlo draws per scenario in calc_probs (default: triceratops default, 1e6). Raise when the log reports that the evidence is dominated by one run.
Do not pool runs	`--no-pool`	off	Do not pool the scenario evidences over the runs; report the mean of the per-run FPPs instead.
Sensitivity runs	`--sensitivity`	off	Also report FPP/NFPP at several flux-uncertainty scales (extra calc_probs runs).
Runs per sensitivity case	`--sens-runs SENS_RUNS`	3	calc_probs runs per scale for `--sensitivity` (pooled).

GUI field	Command line	Default	Meaning
Model exposure [min]	`--tri-exptime-min M`	–	Exposure smoothing of the TRICERATOPS models [min] (default: bin width + native cadences; 0 = package default of 2 min).
Max. runs	`--tri-max-iter N`	–	Maximum calc_probs runs while not converged (default 3 x `--n-iter`).
Time budget [s]	`--tri-time-budget S`	–	Time budget for the extra runs [s] (default 3600).
TRILEGAL / field-star file	`--trilegal-fname TRILEGAL_FNAME`	–	Pre-downloaded TRILEGAL table for triceratops (skips the SSL-sensitive online query).
TRICERATOPS+	`--tri-plus auto|on|off`	auto	Pass ground light curves to calc_probs when the installed triceratops is TRICERATOPS+ (auto, default), require it (on) or never (off).

Tab: Ground photometry

GUI field	Command line	Default	Meaning
Ground light curves	`--ground-lc SPEC`	–	Ground-based light curve (repeatable): 'path,band=r,star=TIC,ap=4.0, fwhm=2.5,obs=CODE,detrend=poly1, AIRMASS'. band: B V R I g r i z J H K (Rc=R, Sloan r'=r); star: TIC at the aperture centre (default: target); ap: aperture RADIUS [arcsec]; fwhm: seeing [arcsec]; time=BJD_TDB (default) | JD_UTC | HJD_UTC; fcol/ecol/tcol: column names; mag=1 for magnitudes.
Ground list file	`--ground-list FILE`	–	File with one `--ground-lc` specification per line (paths relative to that file).
Do not remove cleared stars	`--ground-no-clear`	off	Report the ground clearance but do not remove stars from TRICERATOPS.
Mid-time window [min]	`--ground-tc-window MIN`	–	Minimum half-width of the ground mid-time search [min] (default 60; widened to 3 sigma of the TESS ephemeris).
Detection S/N	`--ground-min-snr GROUND_MIN_SNR`	–	Detection threshold for ground light curves (default 5).
Depth tolerance	`--ground-depth-tol GROUND_DEPTH_TOL`	–	Fractional tolerance on the expected ground depth (default 0.25).
Default seeing ["]	`--ground-fwhm ARCSEC`	–	Seeing FWHM used when a light curve gives none (default 3.0).

Keys of a ground light-curve specification

In the GUI these are the fields of the dialog. On the command line they are written as `--ground-lc "file,band=R,star=65344134,ap=3.5,fwhm=2.4"`.

Key	Meaning
band	B V R I (Johnson–Cousins; Rc = R), g r i z (Sloan; r', SR, rp accepted), J H K, CBB, CV.
star	TIC at the aperture centre (default: target).
ap, fwhm	Aperture radius and seeing FWHM [arcsec].
obs	Observer.
time	BJD_TDB (default), JD_UTC, HJD_UTC.
tcol, fcol, ecol	Column names of time, flux, error.
mag	1: magnitudes.
detrend	auto, poly0/1/2, none, plus covariate columns.
ra, dec	Aperture centre [deg] instead of a star.
pixscale	[arcsec/pixel], to convert AstroImageJ radii.
plus	Band for TRICERATOPS+ (g r i z J H K).
label	Name in the log and figures.

Chapter 11

Output files

All files are written to the output folder (default `TIC<tic>/`).

File	Content
`triceratops_output_TIC<tic>.txt`	The complete log, ending with the SUMMARY block.
`summary_<tic>.json`	Key results (shown on the tab *Results*).
`tesscut_<tic>_sNN.pdf`, `field_<tic>_sNN.pdf`	Cutout, aperture and field per sector.
`lc_panels_*_<tic>.pdf`, `lc_transits_sectors_<tic>.pdf`	Light curves per sector, transits marked.
`lc_stitched_<tic>.txt`, `lc_*_<tic>_sNN.txt`	Light-curve data.
`tls_diagnostics_<tic>*.pdf`	TLS periodogram and fold (blind search).
`ephem_grid_<tic>.pdf`	(P, T_0) grid search.
`per_transit_depths_<tic>.pdf/.txt`	Depth and mid-time of every transit; odd/even.
`secondary_search_<tic>.pdf/.txt`	Secondary-eclipse search.
`phased_transit_<tic>.pdf`, `folded_lightcurve_<tic>.txt`	Folded light curve.
`ld_fit_<tic>.pdf`, `ld_fit_<tic>_bins.txt`	Limb-darkened fit.
`etv_<tic>.pdf/.txt`	Eclipse timing and depth variations.
`difference_image_<tic>_sNN.pdf`, `difference_imaging_<tic>.csv`	Difference imaging per sector; offsets.
`neighbour_vetting_<tic>.csv`	Neighbours' own light curves (`--vet-neighbours`).
`ground_lightcurves_<tic>.pdf`	Fits of the ground light curves.
`ground_clearance_<tic>.pdf/.csv`	Clearance per light curve and star.
`ground_chromaticity_<tic>.pdf`	Depth per band.
`ground_times_<tic>.txt`	Ground mid-times (BJD_TDB, error in min).
`ground_plus_<tic>_<band>_<n>.txt`	Input files for TRICERATOPS+.
`plot_fits_<tic>.pdf`	Best fit of every TRICERATOPS scenario.
`red_noise_scramble_<tic>.txt`	Red-noise test (if requested).
`field_TIC<tic>.csv`, `trilegal_TIC<tic>.*`	Cached catalogue and field-star tables.

Chapter 12

Troubleshooting

Problem	Solution
GUI does not start: `No module named tkinter`	Use a conda Python, or `sudo apt install python3-tk`.
GUI: "has no build_parser()"	The validation script is older than 4.1.1; use the current version.
Red text under the command	An option is missing or two options exclude each other; the text says which.
TRILEGAL download fails (SSL, 403)	Fill in the TRILEGAL form in a browser, download the table, give it as *TRILEGAL / field-star file*; the pipeline otherwise falls back to Gaia DR3.
`No module named pkg_resources` (TRICERATOPS+)	`pip install "setuptools<81"` in that environment.
"QuadraticModel object has no attribute evaluate_pv", or the run stops with a message about pytransit	triceratops needs pytransit 2.2 (newer versions removed this function): `pip install "pytransit==2.2.0"`. *Check* in the GUI shows the pytransit version.
TRICERATOPS+: "only 0-dimensional arrays can be converted to Python scalars", or the run stops with a message about NumPy 2.4	TRICERATOPS+ needs NumPy < 2.4: `conda install -c conda-forge "numpy<2.4"` in the `triplus` environment. *Check* in the GUI warns when the version is too new.
TESScut slow or failing	Retries are automatic; set a *TESScut cache folder* that is reused by later runs.
Warning: depth passed to TRICERATOPS differs from the light curve	Give the depth on the scale of the light curve: on-target with the default dilution correction, the aperture depth with "none" or TRICERATOPS+.
"Monte Carlo not converged"	The verdict is provisional; raise *Max. runs* or *Time budget*, or *Samples per run*.
Ground light curve "NOT USABLE"	The file does not cover the transit window with baseline: check the time system and the date.
Ground: column not found	Give *Flux column*/*Error column*/*Time column* in the dialog.
Files do not open from *Results* under WSL	Open the output folder in Windows Explorer: `D:\\EXPLORE\\candidates_dymock\\TIC<tic>`.

Chapter 13

Glossary

BJD_TDB / BTJD. Barycentric Julian Date in the TDB time scale. BTJD = BJD_TDB − 2457000 is used by TESS.

T0, P, T14. Mid-time of a reference transit, period, total transit duration (first to fourth contact).

Depth. Fractional flux loss at mid-transit, in ppm (parts per million; 1 ppt = 1000 ppm ≈ 1.09 mmag).

Dilution. Reduction of the observed depth by light of other stars in the aperture. "On-target" depth = the depth the target alone would show.

Grazing. The occulting body covers only the edge of the star: V-shaped dip, $b + k > 1$.

FPP, NFPP. False-positive probability; the part due to resolved nearby stars.

TESScut, FFI. Cutouts of the TESS full-frame images.

TRILEGAL. Model of the stellar population of the Galaxy, used for the number of background stars.

Contrast curve. Detection limit (in magnitudes) of companions as a function of separation, from high-resolution imaging.

ETV / O−C. Eclipse timing variations; observed minus calculated mid-times.

References

- S. Giacalone, C. D. Dressing et al., AJ 161, 24 (2021): TRICERATOPS.
- J. Gómez Barrientos et al., AJ 170, 148 (2025): TRICERATOPS+; https://github.com/JGB276/TRICERATOPS-plus.
- M. Hippke & R. Heller, A&A 623, A39 (2019): Transit Least Squares.
- M. Hippke et al., AJ 158, 143 (2019): wotan.
- K. A. Collins et al., AJ 153, 77 (2017): AstroImageJ.
- M. Riello et al., A&A 649, A3 (2021): Gaia EDR3 photometry.
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Statistical validation of TESS (and Kepler) candidates

TRICERATOPS validation pipeline — Observer's Guide

Kepler / KIC update addendum

Pipeline **triceratops_validation.py v4.2**
GUI **triceratops_gui.py v1.2**
November 2026

This short addendum updates the Observer's Guide (originally written for pipeline v4.1.1 / GUI v1.0) for the Kepler support introduced in v4.2 and for the matching GUI fields. Keep it together with the main guide. The technical manual remains the reference for methods and tests.

1. Version summary

The title page and Chapter 10 of the existing Observer's Guide still say pipeline v4.1.1 and GUI v1.0. The code you are using is:

Component	Version	Main change relevant to observers
triceratops_validation.py	4.2	Kepler mode, host-star options, stellar parameters
triceratops_gui.py	1.2	Reads the live parser; shows the new fields automatically
triceratops_audit.py	0.1	Unchanged (prototype audit layer)

The GUI version appears in Help → About. The pipeline version is written into every run log and into `summary_<TIC>.json`.

2. What Kepler mode does

From v4.2 the same pipeline can validate long-period Kepler candidates. Set `--mission kepler` and give the KIC number. The script then:

- Loads Kepler long-cadence SAP / PDCSAP light curves and target-pixel files **per quarter** (quarters take the place of TESS sectors).
- Works in **BKJD** (BJD_TDB – 2454833) instead of BTJD.
- Uses Kepler-band limb darkening.
- Calls TRICERATOPS with `mission="Kepler"` (4″ pixels).
- Looks up the corresponding TIC number from the KIC when `--tic` is omitted (with careful handling of ARTIFACT / DUPLICATE / SPLIT flags).
- Writes results by default into a folder named `KIC<kic>/` (instead of `TIC<tic>/`).

TESS remains the default. Nothing changes for existing TESS workflows if you leave `--mission` at its default value `tess`.

3. New options (Target & ephemeris tab)

These five options were added in v4.2. They already appear in the GUI (the GUI reads `build_parser()` from the validation script). They were missing from Chapter 10 of the v4.1.1 Observer's Guide.

3.1 Command-line reference

GUI field	Command line	Default	Meaning
Mission (TESS / Kepler)	--mission tess|kepler	tess	tess (default) or kepler. With kepler the pipeline uses long-cadence Kepler light curves and pixel files per quarter; times are BKJD (BJD – 2454833); requires --kic.
KIC ID (Kepler)	--kic N	—	Kepler Input Catalog number. Required when --mission kepler. The TIC is looked up automatically if --tic is not given.
Host star for TRICERATOPS (TIC)	--host-tic TIC	—	Star that TRICERATOPS treats as the host (e.g. a close companion). The light curve and pixels stay those of --tic / --kic. The light curve is then used on the aperture scale (SAP, no dilution correction).
Stellar parameters from	--star-source tic|kepler	tic	Host-star R, M, Teff, logg: tic (default) or kepler (Berger et al. 2020 Gaia-Kepler catalogue, else the Kepler DR25 stellar table; needs --kic).
Manual R M Teff [logg]	--stellar R M Teff [logg]	—	Manual host-star parameters R [R■] M [M■] Teff [K] [logg]. Overrides --star-source. Give 3 or 4 numbers.

3.2 Typical Kepler command

Example for a known Kepler ephemeris (numbers are illustrative):

```
python triceratops_validation.py \
--mission kepler --kic 3755522 \
--ephemeris 10.5 245.3 1200 3.2 \
--ld-fit --outdir KIC3755522
```

T0 may be given as BKJD or as the full BJD_TDB; the script converts a full BJD automatically. Depth is in ppm, duration in hours — same convention as for TESS.

3.3 GUI behaviour

On the **Target & ephemeris** tab you will now see the five fields above (Mission, KIC ID, Host star, Stellar parameters from, Manual R M Teff). When Mission = kepler, set the KIC ID; the TIC field can be left empty (the pipeline resolves it). Hover over any label for the full help text that comes from the validation script.

4. Practical notes for Kepler targets

- **Output folder.** Default is `KIC<kic>/`. You can still override with `--outdir`.
- **Time system.** All internal times and the ephemeris T0 are BKJD. The GUI "Predicted transit times" tool switches the zero-point automatically when Mission = kepler.
- **Light-curve source.** For Kepler the default `tesscut` is switched to `pdcsap` (there are no FFI cutouts). You can still choose SAP if you need the aperture-scale flux (e.g. together with `--host-tic`).
- **Stellar parameters.** Prefer `--star-source kepler` (or a manual `--stellar` row) when the TIC parameters look wrong for an old KIC entry that was later split or duplicated.
- **Ground photometry & TRICERATOPS+.** Unchanged. The same `--ground-lc` / `--ground-list` syntax and the same clearance / chromaticity logic apply.
- **Audit script.** `triceratops_audit.py` works on a finished output folder regardless of mission; pass the folder (and optionally `--tic` if the name is ambiguous).

5. What did not change

Everything else in the Observer's Guide remains valid for both TESS and Kepler runs:

- Installation, conda environments, and starting the GUI (Chapter 5).
- Profiles, command import/export, regression tests (Chapters 6 and 9).
- All options on the Light curve, Shape & tests, Field, TRICERATOPS and Ground photometry tabs (Chapter 10).
- Meaning of FPP / NFPP and the PASSED criteria (Chapter 2).
- How ground-based light curves clear neighbours and feed TRICERATOPS+ (Chapter 4).
- Output file names (Chapter 11) — only the default folder name changes for pure Kepler runs.

6. Checklist for a full reprint of the Observer's Guide

When the main PDF is regenerated, the following minimal edits are enough:

1. Title page: "Pipeline … v4.2 GUI … v1.2".
2. Abstract / introduction: one sentence that the pipeline also supports Kepler targets via --mission kepler --kic.
3. Chapter 10 header: "every option of triceratops_validation.py v4.2".
4. Chapter 10, Target & ephemeris table: insert the five rows from §3.1 of this addendum.
5. Chapter 11: note that the default output folder is KIC<kic>/ when --mission kepler.
6. Optional: refresh Figure 6.1 so the Mission / KIC fields are visible.

End of addendum. For methods, tests and the full option list as implemented in code, see the technical manual and the live `python triceratops_validation.py --help` *(or Help → Command-line help inside the GUI).*